

Murad Smretab Mebrahtu

Abu Dhabi, UAE • +971 56 833 8097 • murdiszm@gmail.com
github.com/Murdism • [Google Scholar](#)

Research Interests

My research focuses on uncertainty-aware, deployment-conscious autonomy across the perception, prediction, and planning stack, with the broader goal of enabling safe, risk-aware navigation through improved perception. I am interested in calibrated 3D perception under multi-modal sensor fusion, and in multi-agent trajectory prediction with explicit uncertainty representations. A central aim is to propagate these uncertainty signals into motion planning, so that decisions degrade gracefully under distribution shift, rare events, and interactive human behavior. Complementary interests include large-scale dataset development for perception, multi-task learning, and robust generalization across different environments and domains.

Education

Khalifa University

Jan 2021 – May 2023

MSc in Computer Science | Abu Dhabi, UAE

GPA: 3.93 / 4.0

- Thesis: *Probabilistic Pedestrian Trajectory Prediction for Autonomous Vehicles*.
- Advisor: Dr. Majid Khonji. Co-advisors: Prof. Khaled Elbassioni and Prof. Jorge Dias.
- Full scholarship; coursework in machine learning, computer vision, advanced algorithms, multi-agent systems.

Khalifa University

Jan 2016 – Dec 2020

BSc in Computer Engineering | Abu Dhabi, UAE

GPA: 3.8 / 4.0, *Highest Honors*

- Full scholarship; senior project on multimodal biometric authentication using machine learning.

Publications

Google Scholar: 79 citations • *h-index: 3* • *i10-index: 2*.

Journal Articles

- [J2] N. A. Madjid, A. Ahmad, **M. Mebrahtu**, Y. Babaa, A. Nasser, S. Malik, B. Hassan, N. Werghi, J. Dias, M. Khonji. “Trajectory Prediction for Autonomous Driving: Progress, Limitations, and Future Directions.” *Information Fusion*, Elsevier, article 103588, 2026.
- [J1] E. Shaklab, A. Karapetyan, A. Sharma, **M. Mebrahtu**, M. Basri, M. Nagy, S. Malik, S. Abbas, J. Dias, M. Khonji. “Toward Fully Autonomous Last-Mile Logistics: A Case Study With a Safety-Boosted Self-Driving Delivery Robot.” *IEEE Access*, vol. 13, pp. 182593–182607, 2025.

Conference Papers

- [C2] A. Ahmad, A. Nasser, N. A. Madjid, **M. Mebrahtu**, B. Hassan, J. Dias, M. Khonji. “Priv-DATA: Privacy-Aware Dataset Anonymization Tool for Autonomous Vehicles Research.” *Advances in Science and Engineering Technology International Conference (ASET)*, 2025.
- [C1] **M. Mebrahtu**, A. Araia, A. Ghebreslasie, J. Dias, M. Khonji. “Transformer-Based Multi-Modal Probabilistic Pedestrian Prediction for Risk-Aware Autonomous Vehicle Navigation.” *21st International Conference on Advanced Robotics (ICAR)*, pp. 652–659, 2023.

Preprints and Manuscripts Under Review

- [P5] Y. Babaa, S. M. Naik, **M. Mebrahtu**, S. Abbas, Y. AlSalami, J. Dias, M. Khonji. “GUARDIAN: A Multi-Layer Autonomous Safety Stack with VLM-Guided, Consensus-Resilient Decision Making.” Under review at *IEEE Access*, 2026.
- [P4] N. A. Madjid, **M. Mebrahtu**, Z. Yagudin, B. Hassan, N. Werghi, J. Dias, et al. “Collaborative Trajectory Prediction via Late Fusion.” Under revision at *Information Fusion*, Elsevier (arXiv:2604.22973), 2026.
- [P3] Z. Yagudin, **M. Mebrahtu**, R. Jin, J. Huang, Y. Yue, D. Tsetserukou, J. Dias, et al. “EagleVision: A Multi-Task Benchmark for Cross-Domain Perception in High-Speed Autonomous Racing.” Under revision at *IROS 2026 arXiv:2604.11400*, 2026.
- [P2] A. Nasser, Y. Baba’a, **M. Mebrahtu**, N. A. Madjid, J. Dias, M. Khonji. “Visual Prompt Based Reasoning for Offroad Mapping using Multimodal LLMs.” *arXiv:2604.04564*, 2026.
- [P1] N. A. Madjid*, **M. Mebrahtu***, A. Ahmad, A. Nasser, B. Hassan, N. Werghi, J. Dias, M. Khonji. “EMT: A Visual Multi-Task Benchmark Dataset for Autonomous Driving in the Arab Gulf Region.” *arXiv:2502.19260*, 2025.

Thesis

- [T1] **M. S. Mebrahtu**. “Probabilistic Pedestrian Trajectory Prediction for Autonomous Vehicles in Crowded Environments.” MSc Thesis, Khalifa University, 2023.

Research Experience

Autonomous Vehicle Lab (AVLab), Khalifa University
Research Engineer

Jun 2023 – Present
Abu Dhabi, UAE

Supervisor: Dr. Majid Khonji. AVLab operates under the Khalifa University Center for Autonomous Robotic Systems (KUCARS).

- **Uncertainty-aware perception and prediction.** Built and maintained a ROS2 perception stack integrating Ouster LiDAR and ZED2 stereo camera for 2D and 3D detection, multi-object tracking, and probabilistic trajectory prediction. Prediction uncertainty is preserved through to the planning interface, which directly motivated the uncertainty-propagation focus of my proposed PhD research.
- **Custom 3D detection via camera–LiDAR fusion.** Designed a 3D detector that lifts YOLO 2D detections into 3D using camera–LiDAR fusion to address the limited pedestrian performance of standard LiDAR-only detectors; validated on ROS bags and in Carla and Gazebo simulations.
- **EMT multi-task driving benchmark (co-lead).** Co-led development of a 570,000+ annotated bounding-box benchmark in Gulf-region traffic; owned annotation, verification, and privacy-preserving anonymization, and trained and evaluated the 2D detection, tracking, and prediction baselines released alongside the dataset.
- **Privacy-aware dataset infrastructure (Priv-DATA).** Contributed to an internal framework for validating and anonymizing autonomous-vehicle datasets, including license-plate and face detection-and-masking components and integrity checks.
- **Real-world validation** Planned and executed structured field-test sessions across urban and closed-track environments; collected, organized, and analyzed 500+ GB of ROS bag data and used the logs in a closed-loop failure-identification, targeted-annotation, retraining, and evaluation cycle.
- **Real-time deployment.** Converted detection models via PyTorch → ONNX → TensorRT with CUDA acceleration on NVIDIA GPU platforms for real-time inference under vehicle latency and hardware constraints.
- **Abu Dhabi Autonomous Racing League (A2RL).** Contributed to the SuperFormula autonomous

racing team (2nd place, 2025, 250+ km/h), supporting perception, vehicle startup procedures, and resolution of sensor and software error codes during race preparation. This work exposed the cross-domain generalization and latency problems that shape my research interests.

- **Cross-functional integration.** Worked with localization, planning, and control researchers to debug system-level failures and ensure perception outputs met downstream interface requirements; supported integration on multiple platforms including EasyMile and Pixmoving.

Khalifa University

Graduate Research Assistant (MSc Thesis)

Jan 2021 – May 2023

Abu Dhabi, UAE

- **Transformer-based probabilistic trajectory prediction.** Designed a multi-modal transformer architecture producing probabilistic trajectory distributions with explicit uncertainty estimates for pedestrians and vehicles; trained and evaluated on real-world datasets in PyTorch with CUDA acceleration.
- **Risk-aware planning integration.** Converted probabilistic forecasts into occupancy-grid risk maps and integrated them with an autonomous vehicle planning system to enable risk-aware velocity and path selection; validated on a real vehicle.

Awards and Honors

Scholarships and Academic Distinctions

- Full Scholarship, MSc in Computer Science, Khalifa University, 2021–2023.
- Full Scholarship, BSc in Computer Engineering, Khalifa University, 2016–2020.
- President’s List (four times), Khalifa University, 2016–2019.
- Member, Golden Key International Honour Society, since 2020.
- Member, IEEE Eta Kappa Nu (HKN), since 2018.

Research and Competition Awards

- **Second Place**, Dubai World Congress for Self-Driving Transport (Academia Category), USD 100,000 prize, 2023.
- **First Place**, Dubai World Congress for Self-Driving Transport, Ground Vehicles Challenge (Academia Category), USD 100,000 prize, 2021.
- **Second Place**, National Huawei ICT Competition, UAE, 2018.
- Excellence Award, Huawei Middle East Competition (Regional Finals), Shenzhen, China, 2018.

Teaching and Mentorship

Khalifa University

Graduate Teaching Assistant

Jan 2021 – May 2023

Abu Dhabi, UAE

- Lab Instructor for Programming Concepts (C++), MATLAB, and introductory programming courses.
- Designed and delivered lab sessions, held office hours, and graded assignments and exams.
- Mentored undergraduate students on software engineering fundamentals and debugging practices.

Autonomous Vehicle Lab, Khalifa University

Research Mentor

2023 – Present

Abu Dhabi, UAE

- Mentored junior team members and visiting students on perception pipelines, ROS2 development, and dataset workflows; provided code reviews and technical guidance.

Service and Outreach

Data Science Panel Discussion for Eritrean Students

Organizer and Host

July 2021

Abu Dhabi, UAE

- Organized and hosted a panel discussion on data science careers and research, with speakers from Amazon and industry practitioners; engaged 70+ Eritrean students from across the UAE and beyond.

Talks, Presentations, and Exhibitions

- Paper Presenter, *21st International Conference on Advanced Robotics (ICAR)*, 2023.
- Exhibitor, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*.
- Exhibitor, *IEEE International Conference on Intelligent Transportation Systems (ITSC)*.
- Exhibitor, *Unmanned Systems Exhibition (UMEX)* and *SimTEX*, Abu Dhabi, UAE.

Technical Skills

- **Deep Learning and Computer Vision:** PyTorch, transformer architectures, multi-modal and multi-task learning, 2D/3D object detection (YOLO variants, PointPillars, custom camera–LiDAR fusion detector), multi-object tracking, probabilistic trajectory prediction, OpenCV, Open3D.
- **Sensor Fusion and Calibration:** Camera–LiDAR intrinsic and extrinsic calibration, time synchronization, point-cloud processing, spatial-temporal alignment.
- **Robotics:** ROS1, ROS2, RViz, Gazebo, Nav2, Carla, rosbag2.
- **Optimization and Deployment:** ONNX, TensorRT, CUDA acceleration, NVIDIA edge platforms.
- **Programming:** Python, C++, MATLAB.
- **Software Engineering:** Git, Docker, Linux, modular software architecture, large-scale dataset processing, evaluation pipelines.

Languages

English (very good command, IELTS band 7); Tigrigna and Bilen (bilingual proficiency).

References

Dr. Majid Khonji

Associate Professor, Khalifa University, Department of Electrical Engineering and Computer Science
majid.khonji@ku.ac.ae

Direct supervisor at AVLab (June 2023 – present); MSc thesis advisor (2021–2023); Current Supervisor. Co-author on ICAR 2023, the EMT benchmark (arXiv:2502.19260), the Information Fusion 2026 survey, and additional AVLab publications.

Prof. Khaled Elbassioni

Khalifa University, Department of Computer Science
khaled.elbassioni@ku.ac.ae

MSc thesis co-advisor; co-PI on the AVLab project funding my current research. Course instructor for Advanced Algorithms and Multiagent Systems at the graduate level.

Prof. Jorge Dias

Director, Khalifa University Center for Autonomous Robotic Systems (KUCARS)
jorge.dias@ku.ac.ae

MSc thesis co-advisor; Director of KUCARS. Co-author on ICAR 2023, the EMT benchmark, the Information Fusion 2026 survey, and additional AVLab publications.